

DotDot Global: Promoting smart city innovations through blockchain technology on Google Cloud

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)



To address frequent surges on its smart app, DotDot Global leverages the highly scalable infrastructure and security solutions of Google Cloud to guarantee high-s-

About DotDot Global

DotDot Global is a tech startup founded in 2019 offering strategic technology solutions for the quick service restaurant industry, online information dissemination, digital marketing and local industry innovation. Its one-stop quick service restaurant business management platform has 12,000 partner stores and 25 million cumulative orders in Taiwan. DotDot Global also builds smart city apps for local governments to accelerate urban digital transformation.

service stability and security.

Industries: Technology

Location: Taiwan

Google Cloud results

- Ensures data security in a hybrid cloud architecture with Anthos
- Helps quickly build an extensive blockchain network on Compute Engine
- Enhances app development efficiency by system integration through Sheets
- Helps complete the development

Scales traffic in five

(<https://cloud.google.com/firewall/>) Cloud Key



About Leyun

Founded in 2016, Leyun Inc., stands from customers' perspective to provide the best solutions with more than 20 years of experience in information security and telecommunications services. Dedicated to building a new information security cloud services ecosystem, Leyun's team is familiar with various information security cloud services, and good at integrating various resources. To keep customers informed of the latest security developments information and solutions, Leyun updates its product information regularly to keep pace with the times.

Google Cloud (<https://cloud.google.com/>)

Artifact Registry

(<https://cloud.google.com/artifact-registry>)

Cloud Build (<https://cloud.google.com/build/>)

Cloud CDN (<https://cloud.google.com/cdn/>)

customers' perspective to provide the best
Cloud Firewall (<https://cloud.google.com/firewall>)

of an
innovative
payment
system
within three
months

Cloud Key Management Service

(<https://cloud.google.com/kms/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Google Cloud Armor

(<https://cloud.google.com/armor/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud IAM (<https://cloud.google.com/iam/>)

Secret Manager

(<https://cloud.google.com/secret-manager/>)

Google Workspace

(<https://workspace.google.com/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

In Taiwan, local governments have been implementing smart city strategies to improve public services and welfare to its citizens. By leveraging digital applications to facilitate communication, using AI to predict floods, to level up urban infrastructure, to reduce air pollution, local officials are seeking ways to maximize the benefits of digital transformation by creatively applying the latest technologies.

Supporting Taiwanese local governments carry out their smart city plans with its tech expertise, [DotDot Global](https://dotdot.cc/en/) (<https://dotdot.cc/en/>) was founded in 2019 as the tech startup that offers strategic technology solutions for the quick service restaurant industry, online information dissemination, digital marketing and local industry innovation. Its one-stop quick service restaurant business management platform, which covers all operation aspects from online payment and food ordering, logistics to enterprise resource planning, currently has 12,000 partner stores and 25 million cumulative orders. Following its success in digitizing the quick service restaurant industry, DotDot Global decided to expand its business to the public sector by partnering with local governments to promote urban digital transformation.

"DotDot Global's mission is to help organizations bring better services to end users through technology, and this applies to both private and public sectors," explains Ting Hsieh, CEO of DotDot Global. "After helping thousands of quick service restaurants improve their operational efficiency, we want to extend our influence and contribute to the digitalization process of local governments in Taiwan."

Keeping citizens informed on a single app with data security compliance

In 2019, the government of Taitung County in eastern Taiwan planned to launch a smart city application named TTPush

(https://play.google.com/store/apps/details?id=com.taitung.ttpush&hl=zh_TW&gl=US)

that can disseminate information to its citizens in a more efficient and effective way. To make citizens aware of new policies, local governments in Taiwan traditionally need to hold physical events and give away little gifts like pens and keychains to attract people's attention. This method proved to be costly, without any real way to measure how widely the information is spread, or received by the masses. Adopting TTPush, the Taitung County government could track how its pushed messages are received. The app uses a blockchain-based system to generate TTCoin, a digital currency that would be used to reward users who have completed surveys on TTPush and could be exchanged for products at local partner stores.

DotDot Global was commissioned by the Taitung County government to develop this innovative app. To ensure that TTPush is highly scalable to provide stable services, the two parties decided to deploy the app in a hybrid cloud environment that includes a public cloud platform with abundant computing resources. At the same time, they needed to comply with the Taiwanese data security regulations by storing all of its data in Taiwan. Running TTPush on Google Cloud (<https://cloud.google.com/>) was the perfect solution because it offers cloud infrastructure with high scalability, comprehensive security mechanisms for hybrid cloud, and has a data center in Taiwan.

"In Taiwan, governmental digital services are mostly deployed in on-prem environments to be compliant with the Taiwanese data security rules, but the limited scalability is sometimes detrimental to user experience," explains Hsieh. "With its data center in Taiwan, Google Cloud enables us to offer highly stable services while guaranteeing security."

"In Taiwan,
governmental
digital services
are mostly
deployed in on-
prem

environments to
be compliant
with the
Taiwanese data
security rules,
but the limited
scalability is
sometimes
detrimental to
user experience.

With its data
center in Taiwan,
Google Cloud
enables us to
offer highly
stable services
while
guaranteeing
security."

—Ting Hsieh, CEO, DotDot
Global

Smoothly tackling traffic surges with GKE and Cloud CDN

One advantage that the DotDot Global team
immediately noticed after adopting Google Cloud is
the timely technical support. Since the app

development stage, the technical team of Google Cloud and its partner Leyun (<https://www.leyun.cloud/>) have always been providing instant and detailed answers to all the challenges that DotDot Global encountered. Hsieh says that it's extremely helpful for the DotDot Global team to be able to find technical solutions in a short time, because it can more easily ensure good service quality, which is a primary requirement for a governmental service like TTPush. As a result, TTPush has quickly gained popularity since its launch. As of November 2022, more than half of the people living in Taitung Country had become users of the app.

DotDot Global uses Kubernetes microservices on Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) to run TTPush. Since it's quick and easy to scale up microservices on GKE, DotDot Global is able to smoothly tackle the traffic surges that emerge when the Taitung County government launches a new event or starts giving away TTCoins on the app. In 2021, the county government used TTPush as the platform for citizens to register for COVID-19 vaccines, but the instant traffic spike after registration opened caused the app to crash. With GKE, the DotDot Global team can now scale up microservices by 36x and put the service back into operation in only five minutes. To further improve the user experience, DotDot Global also leverages Cloud CDN (<https://cloud.google.com/cdn>) to speed up content delivery.

"For an application that provides governmental services, maintaining high service availability is

particularly crucial. With Google Kubernetes Engine providing highly scalable microservices and Cloud CDN supporting fast content delivery, we're able to ensure excellent user experience despite traffic surges," notes Hsieh.

"For an application that provides governmental services, maintaining high service availability is particularly crucial. With Google Kubernetes Engine providing highly scalable microservices and Cloud CDN supporting fast content delivery, we're able to ensure excellent

user experience
despite any
traffic surges."

—Ting Hsieh, CEO, DotDot
Global

Guaranteeing security in hybrid cloud with Anthos

DotDot Global adopts a hybrid cloud architecture for TTPush, with some app data stored on-prem. To safeguard citizens' data privacy, the startup employs Anthos (<https://cloud.google.com/anthos>) to efficiently implement unified security policies across on-prem and cloud services, deploy zero trust security models, and ensure all communications to and from TTPush are encrypted. With Anthos, DotDot Global can also easily make sure that the whole hybrid cloud environment is compliant with the international security requirements that the Google Cloud infrastructure meets, such as PCI and ISO/IEC 27001.

"Data security is a primary concern for governments running smart city applications," notes Hsieh.

"Having a hybrid cloud architecture makes it more challenging to ensure the same level of security on both the on-prem and cloud. With all the security mechanisms offered by Anthos, we can achieve high-standard security effortlessly."

Enhancing development efficiency for blockchain-based services and applications

TTCoin is the first digital currency issued by a local government in Taiwan. As users can buy products at local stores with TTCoins, the currency not only encourages citizens' engagement, but also helps boost the local economy. To avoid alterations to the coin, DotDot Global leverages the blockchain technology to deploy TTCoin, with all the currency records stored in every node of the blockchain network. The higher the number of the blockchain nodes is, the more safeguarded the records are. Hsieh says that setting up all the nodes could have been a time-consuming process, but with numerous virtual machines on Compute Engine (<https://cloud.google.com/compute>) available to be used as servers, DotDot Global was able to complete building an extensive blockchain network in a short period of time.

The startup also adopts Google Workspace (<https://workspace.google.com/>) to accelerate product development. Before, to launch a service, DotDot Global's R&D team needed to write scripts in Excel files and then send them to backend engineers for deployment, which required a lot of back-and-forth communication during the process. Now, the team writes all the codes on Sheets (<https://www.google.com/sheets/about/>), which is integrated with DotDot Global's systems and can directly apply the updated scripts. Since all the edits are synchronized instantly online, the time required

for communication has been greatly reduced, and the team is able to launch new services faster.

Swiftly developing an innovative payment service with cloud tools and the local data center

In 2023, DotDot Global decided to upgrade the integrated payment flow service of its restaurant business management platform, so that restaurants can better manage their finances and enhance payment collection efficiency. The company planned to release an innovative payment service that combines online and offline non-cash payments, which not only supports mobile and credit card payments, but also incorporates digital currencies issued by Taiwanese local governments like TTCoin to further promote sustainable digital governance.

With its excellent experience of using Google Cloud tools, DotDot Global chose to develop and deploy its new payment service in Google Cloud. Hsieh notes that the infrastructure supported by GKE can provide high service stability and scalability, and the data center of Google Cloud in Taiwan enables financial service providers to quickly acquire related certifications, such as Payment Card Industry Data Security Standard (PCI DSS). On top of that, DotDot Global leverages Google Cloud Armor

(<https://cloud.google.com/armor>) to further ensure security for payment services by preventing network attacks, and uses Identity and Access Management (<https://cloud.google.com/iam>) to realize access control for the payment system based on Google Workspace accounts. This way, DotDot Global can

make sure that its internal personnel change would not lead to any information security issues.

Google Cloud has also helped DotDot Global accelerate its product development. Not only can the API authentication service of Anthos simplify the process of API authorization, DotDot Global's development team also centralizes the management of software kits that are under development through Artifact Registry

(<https://cloud.google.com/artifact-registry>), which can be easily integrated into the CI/CD workflow in Cloud Build (<https://cloud.google.com/build>). As a result, its software deployment is highly automated, and the development efficiency is enhanced.

In terms of security, DotDot Global adopts Cloud Firewall (<https://cloud.google.com/firewall>) as the firewall of its new payment service. Compared with on-prem firewalls, Cloud Firewall supports easier, faster installation and settings, and can effectively separate networks of different projects, which enables the team to flexibly manage online payment traffic. On top of that, DotDot Global stores sensitive data with Secret Manager

(<https://cloud.google.com/secret-manager>) and manages encrypted keys with Cloud Key Management

(<https://cloud.google.com/security-key-management>), which shortens the time to acquire the PCI DSS certification from 6-12 months to three months. Within less than three months, DotDot Global swiftly completed the development of its new service and launched DDPay in Taiwan in July 2023.

"Our new payment platform is the first one that adopts the Payment-as-a-Service model, which can help restaurants enhance their payment collection efficiency and service quality through remarketing based on payment data," says Hsieh. "The cloud tools of Google Cloud not only provide excellent infrastructure stability and network security, but also help us greatly enhance development efficiency to quickly bring our innovative product to the market."

Expanding smart city features on Google Cloud

Moving forward, DotDot Global plans to leverage the solid IT architecture built on Google Cloud to help the Taitung County government launch more features on TTPush, such as non-profit event promotion and using TTCoins as rewards to encourage citizens to do sports. Following the success of TTPush, DotDot Global has developed a similar platform on Google Cloud for the government of Taoyuan City in northern Taiwan and migrated its food ordering services to Google Cloud. The company also plans to launch DDPay in overseas markets, aiming at reaching a total payment amount of tens of billions of Taiwanese dollars on the payment platform in 2024.

Hsieh says, "Leveraging Google Cloud, we successfully created a smart city app best practice that offers quality and secure services. We believe that with the advanced technologies of Google Cloud, we can continue expanding this model by adding more innovative features."

"Leveraging Google Cloud, we successfully created a smart city app best practice that offers quality and secure services. We believe that with the advanced technologies of Google Cloud, we can continue expanding this model by adding more innovative features."

—Ting Hsieh, CEO, DotDot Global